

Least Variance based Modeling of Heart Disease Prediction System using Ensemble Technique

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Abstract— The cardiovascular disease is the leading cause of death worldwide. To reduce the amount of deaths from heart disease, there must be quick and efficient methods for predicting early heart disease. Machine learning algorithms and approaches have been used on a variety of medical datasets to automate the examination of vast amounts of complex data. The research aims to predict whether a patient has heart disease or not. We employed Decision Trees (DT), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Logistic Regression (LR), and Naive Bayes (NB) as basis classifiers in our work. We used a variety of machine learning models using feature selection, hyper-parameter tuning, and ensembling to increase the prediction accuracy. A single model typically has a high variance or bias. Multiple classifiers' opinions could reduce a single model's variance. We used feature selection techniques like Chi-Square (CS) and Mutual Information (MI) along with dimension reduction techniques such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) to identify the best features. Different datasets related to heart disease were used to evaluate our approach. According to the results, the suggested strategy performs better than the current ones.

Index Terms—heart disease, classifiers, feature selection methods, ensemble learning methods.

I. INTRODUCTION

A *disease* is an abnormal medical state that occurs in the human body. In the last 15 years, cardiovascular illness, especially heart attacks and strokes, have been the leading cause of death worldwide, with an estimated 17 million deaths annually, as reported by the World Health Organization (WHO) [1], [2]. The Institute for Health Metrics and Evaluation (IHME) survey indicates that the death rate from heart attacks rose by 41% between 1990 and 2013. In 2014 alone, the rate shows a significant increase of 9.83% [3].

Early identification is vital for efficient treatment and prevention of heart disease. Predictive algorithms can pinpoint those most at risk for cardiovascular disease, paving the way for early intervention in the form of lifestyle modifications or medications [4–6]. In a healthcare system with limited resources, cardiac disease prediction can improve efficiency.

This allows healthcare providers to concentrate on those patients who pose the greatest threat to the healthcare system. Predictive models can be used for real-time patient monitoring, with alarms and suggestions provided as the patient's risk level fluctuates.

A person's genetics, high blood pressure, poor diet, high calcium rate, constricted blood vessels, etc. are major causes of heart disease [7], [8]. Heart disease is a serious illness that impairs the performance of the heart and can lead to consequences such as infection of the coronary artery and a reduction in the blood vessel's ability to perform its role [9]. Patients with heart disease often do not experience symptoms until the disease has progressed significantly [10], [11].

The use of statistics and modeling in conjunction with data from the present and the past to make projections about the future performance of an organization is referred to as predictive analytic [12], [13]. Data, statistical algorithms, and machine-learning techniques are used in predictive analytic, which is the process of predicting future outcomes based on historical data. Extraction of information from already-existing data for the purpose of identifying patterns and forecasting future events is what predictive analytic refers to [14]. In order to create accurate forecasts, it applies a variety of methods, such as data mining, statistical modeling, and machine learning [15].

Over the past several years, coronary heart disease has emerged as the top cause of death around the globe [10], [16], [17]. There are many different sorts of causes, reasons, and factors that contribute to an increased risk of developing heart disease [18]. Heart attacks can be caused by a number of different diseases. Diseases such as coronary heart disease [15], angina heart disease [26], congestive heart failure [25], congenital heart disease [8], arrhythmias [4], heart attack [25], heart cancer, and many others are taken into consideration in this disease, which is particularly specific and very dangerous to cardiovascular disease or coronary heart disease [19].

For a variety of reasons, human inspection of big data sets and manual medical diagnosis are challenging. Machine learning and artificial intelligence can process data quickly and automate data analysis jobs [20]. This makes it possible to process enormous amounts of patient data effectively. Prediction algorithms can save costs, save time, and simplify diagnosis. People can greatly benefit from such prediction systems, especially in rural locations with limited access to medical facilities. Numerous academics have suggested a number of machine learning-based methods to look into the early identification of CHD [20–24].

The following is the primary contribution of our work: To minimize the number of features required for efficient model training, we made use of the pertinent features from the heart disease dataset. We used feature selection and dimension reduction methods to find the optimal features. We employed greed search to identify the optimal model hyper parameters in order to enhance the model's performance. Additionally, we used the ensemble method to develop a meta-classifier in order to reduce the impact of model variance.

The remainder of this paper is organized as follows: In Section II, we reviewed and summarized relevant research. Section III provides a brief overview of the datasets used in the experiment, as well as performance measures. The proposed approach is explained in Section IV. We employed various feature reduction approaches to optimize model parameters using greed search. Section V elaborates on the results and debate. Section VI is where the work concludes. It also specifies the future scope of the endeavor.

II. RELATED WORK

A substantial number of studies have been published on the factors that increase people's risk of acquiring heart disease. Several machine learning algorithms have been studied for their possible application in predicting cardiovascular disease. In this section, we imposed on a survey of promising research in the field of heart disease prediction.

C.T. Ha et al. [5] studied newly discovered risk factors that are connected with death from heart-related causes. They hypothesized that factors such as age, gender, racial makeup, economic status, diabetes-related findings, and body mass index were linked to cardiac-related mortality. In order to evaluate the efficacy of several supervised machine learning algorithms, M.M. Ali et al. [27] used several. These included Multilayer Perceptron, K-Nearest Neighbour (KNN), Adaboost Classifier, Decision Tree, Logistic Regression, and Random Forest. This hybrid approach was proposed by Jebakumar et al. [6], who used a genetic algorithm for feature selection in conjunction with recursive feature elimination. Mohammad and Fahad [23] presented a novel IoMT cloud based heart disease prediction system which uses Modified Salp Swarm Optimization (MSSO) and Adaptive Neuro-Fuzzy Inference System (ANFIS). To avoid the local minima of ANFIS, authors used Leavy flight algorithm to find optimal parameters. Bizimana et al. presented ML based prediction method which combines data scaling, split ratios and best parameters [20]. Farman et al. proposed a deep learning and feature

fusion-based cardiac disease prediction system [28]. Feature fusion combines electronic medical data with extracted features. Relevant features are chosen based on information obtained. Conditional probability determines the weights for each class.

This study of Mienye et al. provides an improved machine learning approach for predicting the risk of cardiovascular disease [29]. The dataset is divided into smaller portions at random using a mean-based splitting algorithm. The multiple CART models are combined into a homogeneous ensemble using an accuracy-based weighted aging classifier ensemble. Gao et al. [7] presented ensemble based approach for CHD prediction. Authors used dimension reduction methods like LDA and PCA for feature selection. Top seven and six features are selected based on score. DT based bagging method outperforms the individually tested ML models. Javeed et al. [2] used optimized random forest classifier for heart failure prediction. Random Search Algorithm (RSA) was used to find the optimal features, and grid search was used to fine tune the model parameters. Some of the recent works are summarized in Table I.

TABLE I. SUMMARY OF STATE-OF-THE-ART METHODS

Literature	Classifiers	Summary	Limitations	Max. Acc. (%)
[30]	Boosting and stacking	Large dataset has been tested using ensemble and cross validation	Tested with raw features and default model parameters	75.10
[10]	KNN, SVM, RF, DT	Various parameter adjustment has been performed to find optimal ones	All traditional ML models are used without much contribution	86.20
[31]	DT	Designed wrist band with high accuracy	Sensitive to over fitting	73.00
[32]	LR, SVM, KNN, RF, NB	Optimal feature selection is done using PCA of 5 ML models	Tested with raw features and default model parameters	73.00
[33]	SVM, KNN, DT, ANN	Feature selection is done based on statistical property of dataset	Tested with raw features and default model parameters	85.00
[21]	GA-ANN, LR, DT, RF, SVM, KNN	Optimal parameters are selected using GA	Low accuracy because of lack of feature selection	73.43
[34]	ECG signals with QRS complex	ECG signals used for predicting heart disease accurately	Quite high training time	82.50
[35]	LR, J48-SVC, RS-DT, NB, NNMLP, KNN	Used bagging method to address issue of single classifier	No feature selection of parameter tuning employed	74.90

III. DATASETS AND PERFORMANCE METRICS

The heart disease datasets is collected from four different repository. Each heart disease datasets contains different instances or record of the patient heart disease data. All Datasets have 12 or 13 features attribute for disease prediction [36]. Some of the datasets have missing values. All datasets contain multivariate characters such as integer, float and continues. Some of the datasets have null values and duplicated values. The target values which are in form of 0 or 1, present or absent. Dataset used in experiment are summarized Table II.

TABLE II. DATASETS USED IN PRESENTED WORK

Name	Attributes	Type	Instances	Positive Instances	Negative Instances	Missing values?
Heart disease [37]	13	Categorical, Integer, Real	1189	629	560	Yes
Heart disease data [38]	12	Categorical, Integer, continues	1024	526	498	No
Clinical heart disease [39]	12	Integer, Float	302	165	137	No
Heart disease dataset [40]	12	Categorical, Real	269	150	119	No

Numerous performance metrics, including sensitivity, accuracy, precision, and F1 score, are used to thoroughly assess the model performance. The following are the mathematical representation of performance measures used in our work:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Here, TP, TN, FP, FN indicates True Positive, True Negative, False Positive and False Negative samples.

IV. METHODOLOGY

The suggested system's block diagram is portrayed in Figure 1. Selecting the most pertinent features can frequently enhance a machine learning model's accuracy and performance. The ability of the model to learn may be limited by repetitive or unnecessary characteristics that add noise to the input. The initial step in the feature selection process involves extracting the raw features from the specific structured data, mutual information, or information gain. We obtain several highly linked features that are more crucial to the prediction of heart disease when we combine the chi squared feature selection technique with mutual information.

The likelihood of the model overfitting to the training set is decreased with fewer features. When a model over fits, it fails to recognize the underlying pattern in the training data and instead learns the noise, which results in poor generalization to new data. For the high accuracy of machine learning model, we are also taking feature dimension reduction methods. Linear discriminant analysis and principal component analysis which are more useful method for models. With less features, training gets easier and model becomes more interpretable.

After the reduction feature method, grid search feature selection which is also used for high performance. In the grid search algorithm, some of the features are selected for training and testing parts. For the high-performance quality of the machine learning models we are using hyper parameter tuning. With the hyper parameter tuning models are adjusted hyper parameters values for achieving the overall best classification accuracy

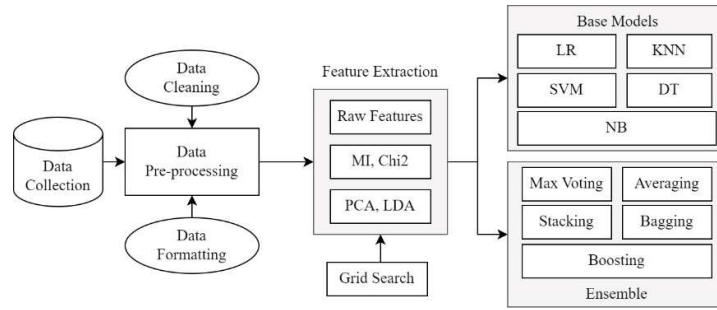


Figure 1. Block diagram of proposed system

For the heart disease prediction machine learning algorithms, proposed system uses logistic regression, k-nearest neighbor, support vector machine, decision tree and naive bytes. Optimal parameter is the key to success of any classifier. To find the best parameters, we used grid search method to find the best pair of parameters. These all models are trained in hyper parameter tuning with the best selected feature selection method. After those tuned models are using for validation and ensemble techniques.

In the validation method simple train-test-split technique which is perform with the performance metrics. And the other ensemble learning max voting, averaging, stacking, bagging and boosting techniques which are used. All the models perform both techniques and gives the accuracy according to performance metrics. Over all the result of all those techniques the boosting algorithms gives the high accuracy compares to other models. We used 10-fold cross validation to measure the generalization ability of the model. The entire dataset is divided in 70-30 ratio. We used 70% data to train the model and 30% to test the model.

V. RESULTS AND DISCUSSION

We evaluated our models under various scenarios. Results of them are discussed here:

A. Comparison of Feature Selection

Many features are encountered in real-world datasets when developing machine learning models, yet not all of these features are always significant. When superfluous features are added during model training, the model's overall accuracy decreases, its complexity rises, its capacity for generalization declines, and it becomes biased [13], [41]. We evaluated each of the five models using both raw and features that were chosen using Chi-Square (CS), Mutual Information (MI), Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA).

The Table III shows the results of all base models for various feature selection approach without tuning hyper parameters of the model. Models are trained with default parameters set by library. It is clearly understood from results that the raw features includes unnecessary attributes for model training, which makes model less general and it face difficulty to establish the association between features and class label. Whereas with feature selection (MI and CS) and dimension reduction (PCA and LDA), we get more relevant features. Results also confirm the usefulness of feature selection and dimension reduction. LDA has better discriminative power as it focus on mean separation of projected data [42].

TABLE III. ACCURACY (%) OF BASE MODELS ON VARIOUS FEATURES WITHOUT HYPER PARAMETER TUNING

Classifier	Raw	MI	CS	PCA	LDA
Logistic Regression	0.60	0.85	0.78	0.87	0.89
K-Nearest Neighbor	0.75	0.84	0.75	0.83	0.84
Support Vector Machines	0.70	0.83	0.76	0.80	0.84
Decision Tree	0.78	0.85	0.85	0.85	0.86
Naive Bayes	0.70	0.84	0.79	0.84	0.86

Model performance is highly influenced by the hyper-parameters of the model. All models have different parameters to tune. We used grid search method to find the optimal set of parameters for each model. The results shown in Table IV with tuned hyper parameter is quite better compared to default parameters as described in Table III. Like without hyper parameter tuning, here also we get the improved results with selected features. With tuned model, accuracy goes as high as 90%.

TABLE IV. ACCURACY (%) OF BASE MODELS ON VARIOUS FEATURES WITH HYPER PARAMETER TUNING

Classifier	Raw	MI	CS	PCA	LDA
Logistic Regression	0.85	0.89	0.88	0.90	0.90
K-Nearest Neighbor	0.82	0.90	0.89	0.89	0.91
Support Vector Machines	0.80	0.88	0.85	0.86	0.87
Decision Tree	0.81	0.90	0.90	0.90	0.91
Naive Bayes	0.84	0.88	0.88	0.87	0.89

B. Comparison of Ensemble Methods

Ensemble technique combines predictions of various models and leverages the prediction power of final classifier [43]. Basically it combines multiple base classifiers to reduce the classification error. It is expected that the results of ensemble model is better than the individual classifier. We used five base models and applied various ensemble techniques like averaging, max voting, stacking, bagging and boosting. Results are portrayed in Table V. Table describes the results of model on default hyper-parameters (HP) and with hyper-parameters. Even without parameter tuning, the ensemble technique shows result as high as 90% for boosting. And with tuned base models, the accuracy of boosting method shoots to as high as 98%. We used Adaboost algorithm for boosting method, Adaboost is a strong classifier derived from series of weak classifiers. This improvement however comes at the cost of model complexity and increased training time.

TABLE V. AVERAGE ACCURACY (%) OF PROPOSED MODEL ON VARIOUS ENSEMBLE METHODS

Classifier	Without HP Tuning	With HP Tuning
Averaging	0.85	0.89
Max Voting	0.82	0.90
Stacking	0.80	0.88
Bagging	0.81	0.90
Boosting	0.84	0.88

Through feature selection and ensemble model testing on dataset 3, we realized that Adaboost with tuned model gives the best accuracy. To check the robustness of proposed method, we tested all four datasets on Adaboost with and without tuned models as base classifiers. The results for all datasets are compared in Figure 2.

C. Comparison with Existing Works

Various machine learning models have been exploited to achieve better accuracy. Although ML models often suffer from bias and variance issues. We tried to address that issue by creating the Meta classifier through

ensemble of various based models. The comparison with state-of-the-art methods shows the superiority of proposed method. We did test our model on four different datasets and model behaves reasonably well on all datasets. AdaBoost model with tuned base model outperforms almost all compared methods on all datasets. Table VI shows the comparison of results of proposed method with existing approaches.

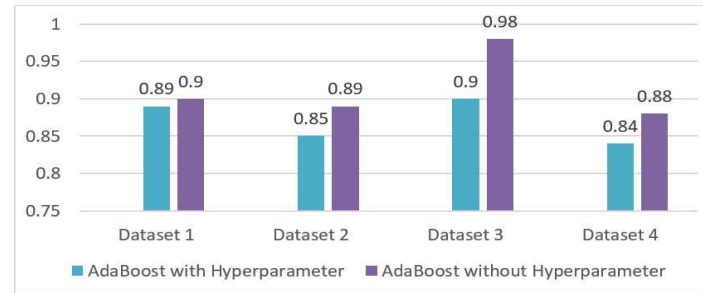


Figure 2. Comparison of proposed method on various datasets

TABLE VI. COMPARISON OF PROPOSED METHOD WITH EXISTING APPROACHES

Literature	DB1	DB2	DB3	DB4
Support Vector Machines [44]	0.83	-	-	-
Random Forest Classifier [45]	-	-	0.87	-
XGBoost [46]	0.87	-	-	-
Logistic Regression [22]	-	-	0.90	-
Logistic Regression [41]	0.78	-	-	-
Support Vector Machines [13]	-	-	0.80	-
Random Forest [24]	0.81	-	-	-
Decision Tree [47]	0.80	-	0.75	-
Proposed (Stacking)	0.92	0.90	0.89	0.85
Proposed (AdaBoost)	0.90	0.98	0.90	0.88

VI. CONCLUSIONS

In this work, we evaluated five machine learning models (i.e. LR, DT, KNN, SVM and NB). We also investigated effect of different feature selection and feature reduction methods (such as MI, CS and PCA) for relevant feature selection. Models are tested without and with parameter tuning to understand the importance of perfectly tuned models. ML models are susceptible to noisy data and tend to over fit. Such models often suffer from high variance problems, especially decision tree. To minimize the variance effect, we used ensemble method, which indeed leverage the performance of model by far margin compared to individual classifier. Adaboost classifier achieves highest accuracy of 98%.

Number of feature vary from dataset to dataset. In future, we are planning to investigate the method which equally effective on all datasets. Also, we are looking forward to explore the way to find the way to select most relevant features from the dataset. Less number of features implies fast learning and more interpretable model.

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